EXP 4

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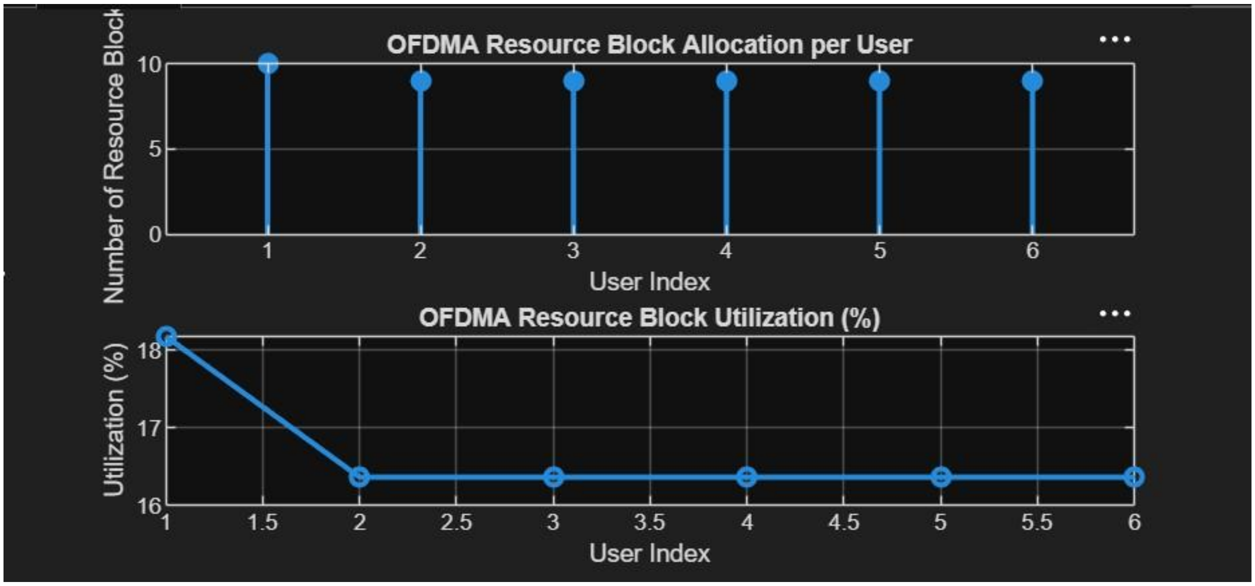
OBJECTIVE 1

CODE:

```
clc; clear; close all ;
% System parameters bandwidth =
10e6; % 10 MHz rb_size = 180e3;
% 180 kHz per RB
% Total Resource Blocks total_RB =
floor(bandwidth / rb_size);
% Users
users = 1:6;
% RB allocation (simple equal + remainder distribution) base = floor(total_RB / length(users));
alloc = base * ones(1,length(users)); alloc(1:mod(total_RB,length(users))) =
alloc(1:mod(total_RB,length(users))) + 1; utilization = (alloc / total_RB) * 100; disp( 'Total
Resource Blocks:' ) disp(total_RB) disp( 'RB Allocation per User:' ) disp(alloc) figure
% -------- Graph 1: RB Allocation per User --------
subplot(2,1,1) stem(users, alloc, 'filled' ,
'LineWidth' , 2) title( 'OFDMA Resource Block
Allocation per User' ) xlabel( 'User Index' ) ylabel(
'Number of Resource Blocks' ) grid on
% -------- Graph 2: RB Utilization --------
subplot(2,1,2) plot(users, utilization, '-o' ,
'LineWidth' , 2) title( 'OFDMA Resource
Block Utilization (%)' ) xlabel( 'User Index' )
ylabel( 'Utilization (%)' ) grid on
```

OUTPUT:

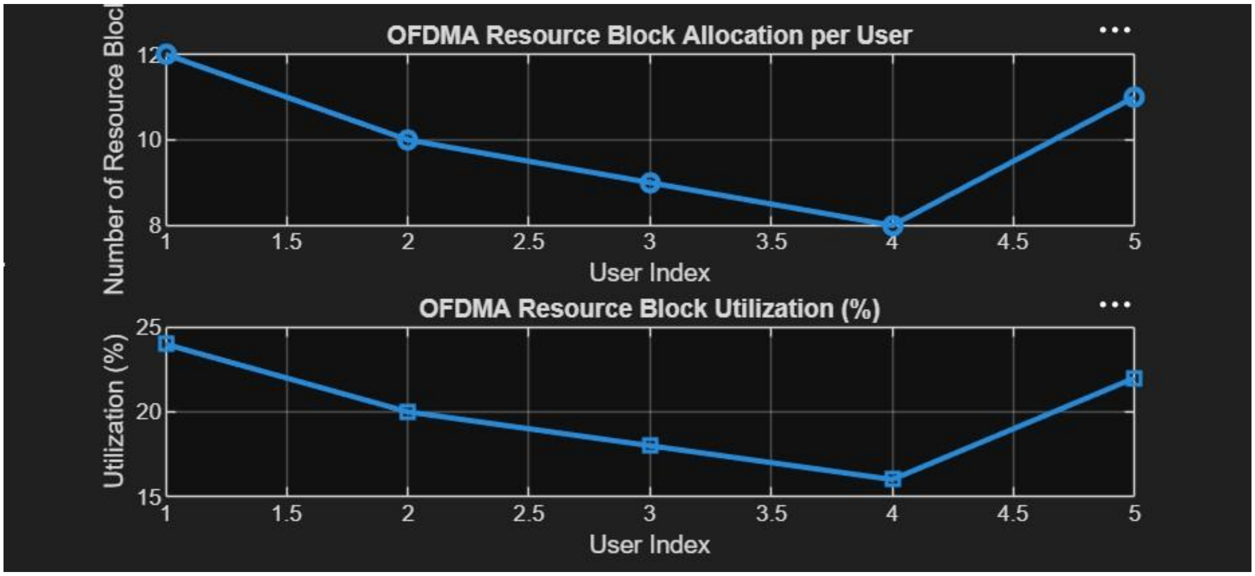


OBJECTIVE 2

CODE:

```
clc; clear; close  all ;
 % Total system resource
blocks  total_RB = 50;  %
Number of active users  users
= 1:5;
 % RB allocation to users (example scenario)
alloc = [12 10 9 8 11];  % Resource Block
Utilization (%)  util = (alloc / total_RB) * 100;
disp( 'Resource Block Utilization (%) per User:'
)  disp(util')  figure
 % -------- Graph 1: RB Allocation --------
subplot(2,1,1)  plot(users, alloc,  '-o' ,  'LineWidth' ,
2)  title( 'OFDMA Resource Block Allocation per
User' )  xlabel( 'User Index' )  ylabel( 'Number of
Resource Blocks' )  grid  on
 % -------- Graph 2: Utilization (%) --------
subplot(2,1,2)
 plot(users, util,  '-s' ,  'LineWidth' , 2)  title(
'OFDMA Resource Block Utilization (%)' )
xlabel( 'User Index' )  ylabel( 'Utilization (%)' )
grid  on
```

OUTPUT:



OBJECTIVE 3

CODE:

```
clc; clear; close  all ;
 % System parameters  rb_bw = 180e3;  %
Bandwidth per RB (180 kHz)  users = 1:5;
 % Allocated Resource Blocks per user  alloc_RB =
[12 10 9 8 11];  % Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2];  % QPSK, 16QAM, 64QAM
etc.
 % Throughput calculation (Mbps)  throughput =
(alloc_RB .* rb_bw .* mod_eff) / 1e6;  disp( 'User
Throughput (Mbps):' )  disp(throughput')  figure
 % -------- Graph 1: Throughput per User --------
subplot(2,1,1)
 plot(users, throughput,  '-o' ,  'LineWidth'
, 2)  title( 'OFDMA User Throughput' )
xlabel( 'User Index' )  ylabel( 'Throughput
(Mbps)' )  grid  on
 % -------- Graph 2: RB vs Throughput --------
subplot(2,1,2)  plot(alloc_RB, throughput,  '-s' ,
'LineWidth' , 2)  title( 'Resource Blocks vs
Throughput Relationship' )  xlabel( 'Allocated
```

```
Resource Blocks' )  ylabel( 'Throughput (Mbps)' )
grid  on
```

OUTPUT: